Bcho NAM Player

Standalone Manual



Standalone | Version 1.8.5

Complete reference for the standalone application on Windows, macOS and Linux. This guide covers the controls, file handling, signal paths and session behavior of version 1.8.5. For the VST3 / AU plug-in see PLUGIN_MANUAL.en.md.

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1. What Bcho NAM Player is

Bcho NAM Player is a portable standalone guitar processor. It plays Neural Amp Modeler captures (.nam), cabinet impulse responses (.wav) and a reorderable chain of studio effects, and it owns the audio device itself: driver, channels, sample rate, buffer size and physical output routing are configured inside the application.

The player has **two complete and independent signal paths**, left and right. In NORMAL, each path owns its own BLOCK NAM 1, BLOCK NAM 2, cabinet IR, effects rack, block order, gate, tone stack, master level, IR blend, IR cabinet volume, input and output gain, power switch, calibration and tuner. In **MONO** only the left path is audible and the right-hand controls are hidden; in **STEREO** both run at once. PLUS expands each path into a series lane or two parallel lanes, each limited to two NAMs and two IRs. Standalone DI playback can replace the physical input (chapter 6).

What is in the package

Item	Notes
Bcho NAM Player.exe (or .app / AppImage)	The whole application; no installer, no runtime DLLs
Models	Two demonstration NAM captures. They are **not** loaded automatically
IRs	The folder the IR browser opens by default. No cabinet IR is bundled
NAM_A2_HEAD.xml	Appears after the first normal exit: your saved session
tone3000.session	Appears only after connecting a TONE3000 account (encrypted)

The window opens centred at its native 1537 x 1023 size whenever the screen allows it. The whole design is responsive - furniture, knobs, lists, meters and text scale together - and cannot be made smaller than 50 % of native size, so no control or legend can ever leave its plate.

2. Installing and first launch

Choosing EmbeddedBrowser or ExternalBrowser

Version 1.8.5 provides two variants for both **Standalone** and **VST3** on Windows x64:

ZIP variant	TONE3000 sign-in and catalogue	Browser requirement
EmbeddedBrowser	Opens sign-in pages and SEARCH FULL CATALOGUE inside the application	Microsoft WebView2
ExternalBrowser	Opens those pages in your default system browser	Does not require WebView2

The TONE3000 library remains available through the TONE3000 button in both variants. The audio engine, controls, NAM models, IRs and presets are the same; only the browser used for sign-in and catalogue selection changes. Local files work offline in both.

Choose a **Standalone** ZIP to run the application, or **VST3** to use it in your DAW. Both VST3 variants share the same plug-in name and identity: install only one at a time. To switch, close the DAW, replace the entire Bcho NAM Player.vst3 folder and rescan plug-ins. Do not combine files from the two packages. You can keep both standalone variants in separate folders. Keep your models, IRs and presets when replacing a variant.

The cover image is a standalone screenshot; the plug-in manual describes the controls available inside a DAW. Embedded browser examples refer to EmbeddedBrowser; in ExternalBrowser, complete those steps in your system browser.

Download the package for your system from the latest public release, extract it, and keep every file and folder together.

System	How to run it
Windows x64	Run Bcho NAM Player.exe. Install your interface manufacturer's ASIO driver for the lowest latency
macOS Apple Silicon	Use the arm64 ZIP, open Bcho NAM Player.app
macOS Intel	Use the x86_64 ZIP, open Bcho NAM Player.app
Linux x86_64	Run the AppImage; chmod +x BchoNAMPlayer-*.AppImage if it will not start

macOS builds are ad-hoc signed but **not** notarized. If Gatekeeper blocks the first launch, right-click the application and choose **Open**.

What the first launch looks like. The player starts in MONO, with every block disabled, no model and no IR loaded, and the audio device set to the system default. A completely bypassed chain passes your dry signal through - it does not mute. Your first job is Audio Setup and then loading a capture.

Every later launch restores the last session you closed normally: controls, both paths, resources, block order, effects and bypass states. The source always returns to IN and DI playback remains paused.

3. Map of the window



The standalone in MONO with a capture, a cabinet and two effects running

The window has three pieces of furniture, from top to bottom. The rack also holds NORMAL / PLUS and IN / DI. CONFIG DI and the gold-framed transport appear only in DI mode; PAN appears in STEREO.

1 - The rack strip. Preset bay on the left, the reorderable processing blocks in the middle, the tuner button, display and CHROMATIC indicator, and on the right the MONO / STEREO switch and the settings gear.

2 - The amplifier head. Input VU, INPUT GAIN and INPUT CALI on the left; the seven main knobs and, above them, the NAM and IR selectors; the two file browsers (NAM models on the left, IR files on the right) with the IR VOL fader between them and the output column; Output VU, OUTPUT GAIN and the red POWER switch on the right.

3 - The speaker cabinet. The Bcho logo and two cones that move with the real output level.



Preset bay, input-mode switch and the first rack blocks

The caption under **LOAD PRESET** always names what the shared controls are editing right now - for example PATH R · NAM 2, or MONO · NAM 1.

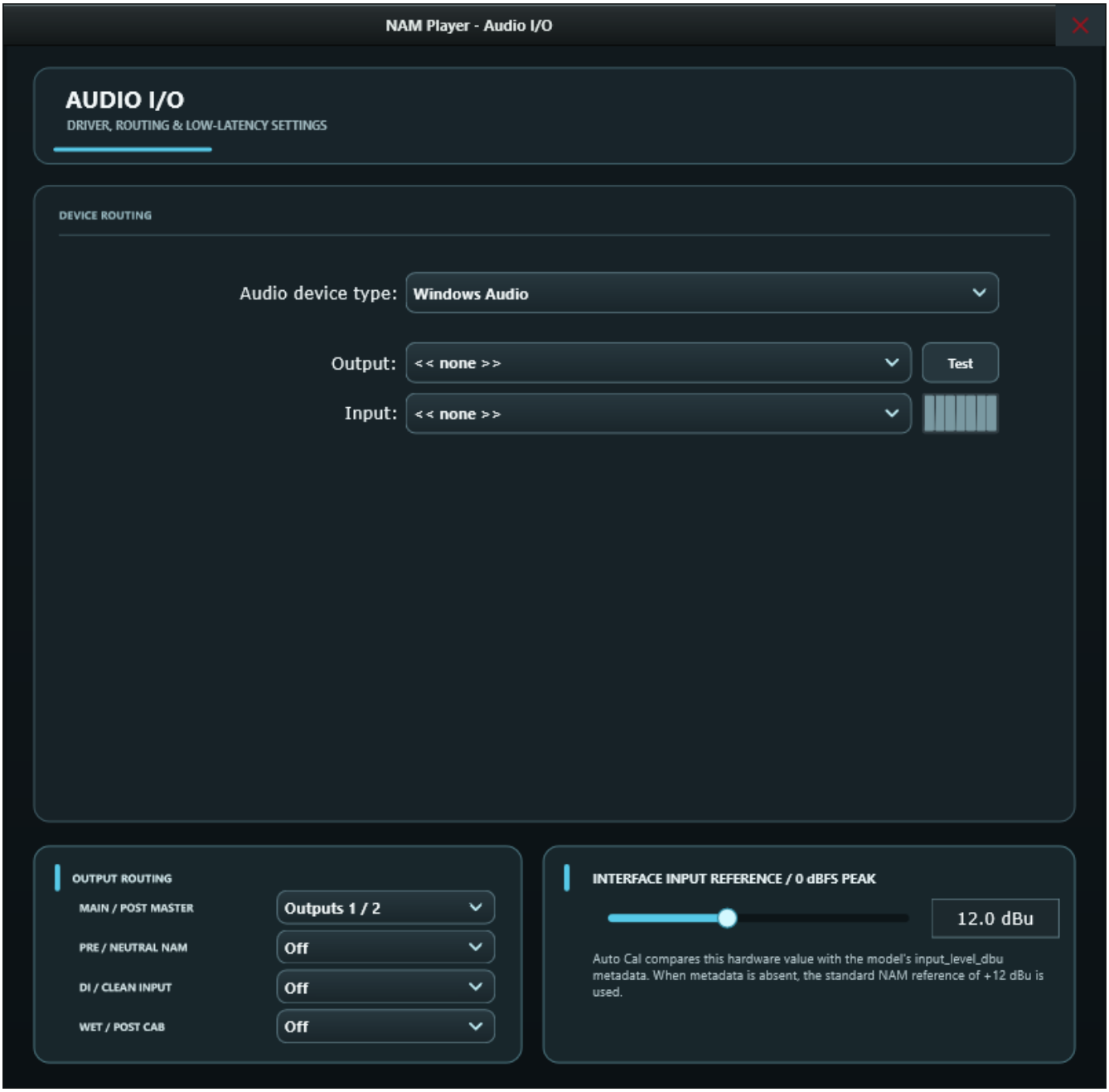
4. Quick start: sound in five steps

1. **Open Audio Setup.** Press the gear (top right of the rack) → **AUDIO SETUP**. Choose your driver and interface, pick the input your guitar is plugged into, set the sample rate and the buffer size, and make sure **MAIN / POST MASTER** points at the outputs your monitors use.
2. **Load an amp capture.** Press **BROWSE LOCAL** under NAM MODELS and pick a .nam file or a whole folder. The capture loads, BLOCK NAM 2 lights up, and the file appears in the list.
3. **Load a cabinet.** Press **BROWSE IR** and pick a .wav impulse response. The IR block lights up. (Skip this if your capture already includes a cabinet - many "amp_cab" captures do.)
4. **Set levels.** Play and watch INPUT VU: aim for the green range with the loudest strumming just touching yellow. Use INPUT GAIN to trim. Then set MASTER VOL and OUTPUT GAIN for a comfortable monitoring level.
5. **Play.** Add effects by clicking their blocks; double-click a block to open its editor.

Start at a low monitoring level and avoid hearing your guitar twice - through the interface's own direct monitoring **and** through the player.

5. Audio Setup: device, latency and routing

Gear → **AUDIO SETUP**.



The Audio I/O window: device routing, output routing and the calibration reference

Device routing (upper half)

Control	What it does
Audio device type	The driver family: ASIO, Windows Audio, DirectSound (Windows); CoreAudio (macOS); ALSA/JACK (Linux). **ASIO gives the lowest latency on Windows**
Output	The interface that plays the processed sound
Input	The interface your guitar is connected to
Active channels	Tick the input channel the guitar uses and the output pair your monitors use
Sample rate	44.1 kHz and up. Higher rates cost more CPU; NAM captures are resampled internally as needed

Control	What it does
Audio buffer size	The latency/stability trade-off. 128 or 256 samples is a good starting point
Control panel / Test	Opens the manufacturer's own panel (ASIO) and plays a test tone

Output routing (lower left). Four independent taps, each assigned to a physical output pair or switched **Off**:

Route	Signal it carries	Typical use
MAIN / POST MASTER	The finished player output	Monitoring and recording the final sound
PRE / NEUTRAL NAM	BLOCK NAM 2 before the player's tone stack, effects and cabinet	Re-amping, A/B comparison
DI / CLEAN INPUT	The selected input source before amplifier processing (physical input on IN, file playback on DI)	A clean safety track
WET / POST CAB	The processed post-cabinet branch	A separate processed recording path

A pair can only be used by one route: choosing a pair that is already taken resets that selector to **Off**. MAIN normally uses outputs 1/2. Leave the routes you do not need switched off so nothing is duplicated.

Interface input reference / 0 dBFS peak (lower right). A slider from 0 to 30 dBu, shown in dBu. It tells Input Cali what your interface's maximum input level is. It has no effect while Input Cali is off.

Everything in this window - device, channels, rate, buffer, reference and the four routes - is saved per machine and restored on the next launch. It is deliberately **not** stored in .bnpp presets, so a preset can travel between computers.

6. Signal flow

IN / DI: choose the source (standalone only)

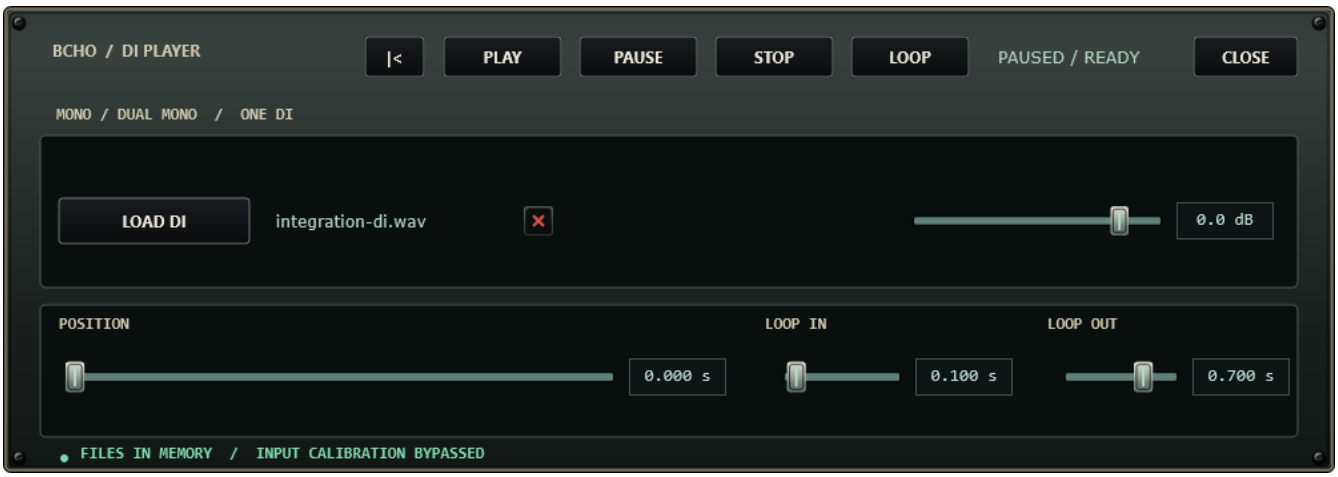
The large **IN / DI** lever sits below DUAL MONO / SPLIT L/R. **IN** uses the configured physical inputs. **DI** replaces them with audio files before the existing effects, NAMs, IRs and output routing. The application starts on **IN** at every launch. Source changes use a short ramp to avoid clicks.

Selecting DI opens the DI PLAYER modal. Close it with **CLOSE**, its window close button or Esc; playback continues. **CONFIG DI** reopens it without changing source. The cabinet and the main browsers stay clear.

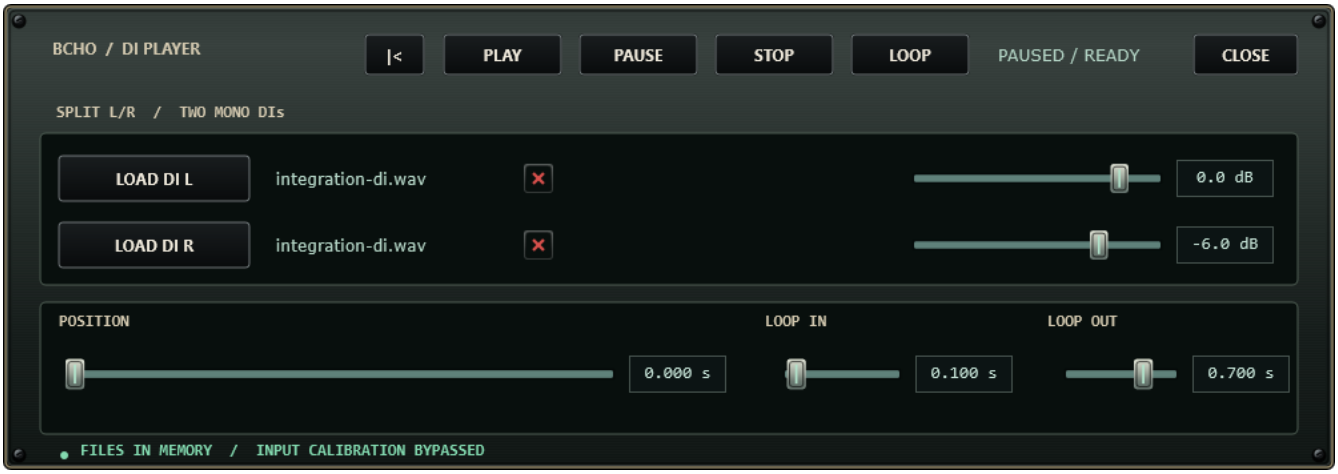


DI source selector and CONFIG DI

Processing mode	DI tracks
MONO	One mono file feeds the active left path
STEREO + DUAL MONO	The same mono file feeds both paths
STEREO + SPLIT L/R	Two mono files, DI L and DI R, independently feed the two paths



The DI player in MONO / DUAL MONO



The DI player in SPLIT L/R

Load and remove. Use **LOAD DI**, **LOAD DI L** or **LOAD DI R**, or drop a file onto its row. In SPLIT you can drop two files together, one per path. WAV, AIFF and FLAC are accepted, **mono only**. A stereo or multichannel file is rejected with a message explaining that each DI must be a mono track; the previous file stays loaded. Use the red **X** next to a filename to unload that track from memory. It does not delete the original file. Removing a track pauses the shared transport and leaves the other track loaded.

Files load in the background and play from memory. Each track may use a different sample rate; playback automatically resamples to the audio device. No automatic normalization is applied. The limit is 512 MiB of decoded audio per track. Each track has its own **-60 to +12 dB** level, initially **0 dB**; double-click its level slider to reset. INPUT GAIN remains available further down the chain.

Shared DI transport and loop

The gold-framed transport above the NAM and IR browsers appears only in DI mode. From left to right: **back 5 seconds, PLAY, PAUSE, STOP, forward 5 seconds**. PLAY lights while playback is running.



The DI transport integrated into the gold-trimmed head

Control in the DI PLAYER	Action
PLAY	Play both tracks from the common position
PAUSE	Keep the position and send silence into the processing paths; effect tails continue
STOP	Pause and return to the beginning
Return-to-start button	Go to the beginning without changing play/pause state
POSITION	Seek with the slider or enter a time in seconds
LOOP	Enable or disable repetition for both tracks
LOOP IN / LOOP OUT	Set the common loop boundaries with sliders or numeric times; while LOOP is on, drag the gold [and] markers on POSITION to move the same boundaries

The timeline lasts as long as the longest loaded track. A shorter or unloaded track supplies silence; it never falls back to the physical input. Both tracks seek and loop together. Closing the modal does not stop the transport.

Switching sources: DI to IN pauses DI and smoothly restores the physical input. Returning to DI recalls its files, gains, position and loop, but **does not start playback**. Press PLAY when ready. The physical input configuration and its calibration setting are retained while DI is selected.

Calibration: interface-specific INPUT CALI is bypassed in DI mode, as shown by the player status. It resumes with its previous setting on IN. This bypass does not alter the per-file DI level or the amplifier's INPUT GAIN.

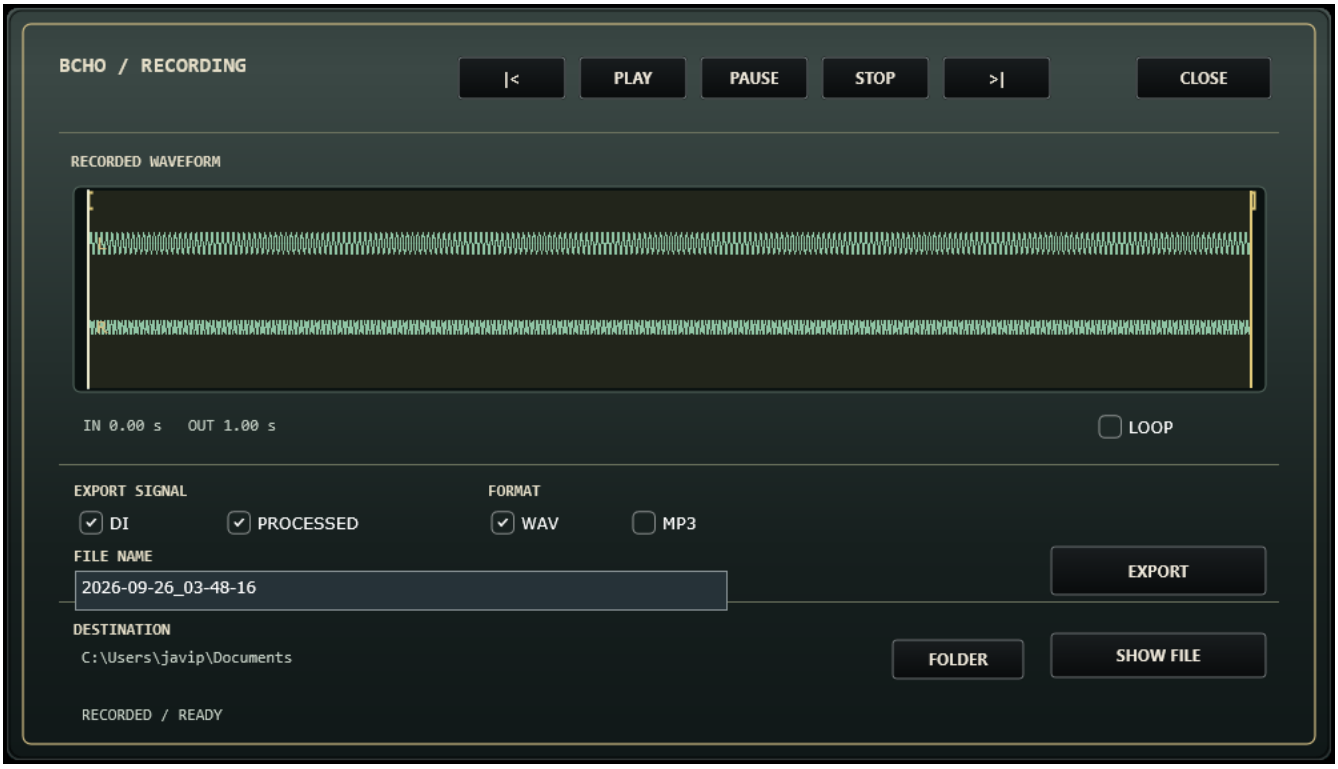
Recall: a normal exit saves file paths, levels, position and loop with the standalone session. On relaunch the source is IN and DI is paused. Keep the files in their saved locations or load them again. DI files and transport settings are excluded from .bnpp presets. The plugin has no DI player, source switch, transport controls, DI parameters or DI playback state; use audio tracks in the DAW.

Recording and exporting a take (standalone only)

In **IN** mode, the gold-framed tolex strip used by the DI transport instead shows ● **REC**. Press it to record the physical input and the final processed MAIN sound at the same time. While recording, the button flashes red and reads ■ **STOP**. Press STOP to finish and open the recording review window. The IN / DI and MONO / STEREO mode switches are locked during a take so its channel layout cannot change halfway through. Recording takes place in the background; disk writing does not run in the audio callback. Changing the audio device ends the take and opens review. Keep enough free disk space.



REC and COUNT-IN on the gold-framed tolex



Recording review and export window

The **DI** recording is captured before INPUT GAIN, INPUT CALI and all effects. In MONO it is one mono signal; STEREO + DUAL MONO stores the shared input as two identical mono channels, and STEREO + SPLIT L/R stores each physical input independently. When exported in STEREO, DI becomes two separate mono files, -DI-L and -DI-R, ready for the DI player. **PROCESSED** is the final MAIN sound after effects, NAM, IR, cabinet positioning, PAN and OUT; it exports as one stereo file in STEREO or one mono file in MONO. Neither recording is normalized automatically. Other optional physical output routes are not mixed into PROCESSED.

The review window displays one waveform in MONO and two lanes in STEREO. **PLAY**, **PAUSE**, **STOP**, **|<** and **>|** audition the processed take, pause it, stop and return to the start, or move back/forward five seconds. Click or drag the white playhead in the waveform to seek. The gold **[** and **]** markers initially enclose the entire take; drag them to choose the section. **LOOP** repeats that section during playback. Export uses exactly the marked section whether LOOP is enabled or not. Review playback goes to the configured MAIN output pair and replaces the live MAIN signal while it plays; closing the review window stops playback.

Select **DI**, **PROCESSED**, or both, and select **WAV**, **MP3**, or both. At least one signal and one format are required. Enter a base filename, or leave it empty to use the current date and time. **FOLDER** chooses the destination. **EXPORT** writes the selected files in the background; the status line lists the result. After a successful export, **SHOW FILE** opens the destination folder with an exported file selected. WAV is 24-bit PCM at the recording device's sample rate. On Windows, MP3 uses the system encoder included with Windows and needs no separate installer; unsupported device sample rates are converted to 48 kHz for MP3. Export does not delete the temporary recording.

Practice and recording metronomes (standalone only)

The three metronome switches have separate jobs. **METRO** beneath the IN source switch turns on the practice click for playing without recording; it does not start REC. **COUNT-IN** beside REC requests a lead-in before the take starts. **CLICK DURING REC** in Settings controls whether beats continue after the lead-in while recording. Each switch is independent: COUNT-IN can be on while METRO and CLICK DURING REC are off. The practice click stops when DI is selected; its switch is replaced by CONFIG DI. All clicks are monitor-only and absent from both DI and PROCESSED exports.



Practice METRO below the IN source selector

Settings shares **BPM** and **TIME SIGNATURE** between the practice and recording metronomes. Defaults are **65 BPM** and **4/4**. Choose 2/4, 3/4, 4/4, 5/4, 6/8 or 7/8. The first beat of each bar is a stronger, higher wooden-stick click. **COUNT-IN BARS** (1-8, initially 1) sets how many whole bars play before any recording samples are written. If COUNT-IN is off, capture starts immediately; if CLICK DURING REC is off, the take runs without click after the lead-in. The metronome is mixed into the listening output only, after both capture points. Tempo, meter, count-in length and recording-click choices are saved in the standalone session, never in plugin presets. Practice METRO does not start automatically on relaunch.

Processing order and output taps

Each path processes its own audio in this order:

selected source: physical IN or DI file(s)
→ tuner tap (always the untouched DI, before everything)
→ INPUT GAIN (+ Input Cali offset only on physical IN)
→ GATE
→ [rack blocks, in the order shown on screen]
...effects before BLOCK NAM 2...
BLOCK NAM 1 (its own BASS/MID/TREBLE/PRESENCE)
BLOCK NAM 2 (then the main BASS/MID/TREBLE/PRESENCE)
...effects between BLOCK NAM 2 and IR...
IR (cabinet convolution → IR VOL → IR BLEND)
...effects after IR...
→ MASTER VOL
→ OUTPUT GAIN
→ POWER
→ MAIN output

Other taps: DI before processing; PRE is a separate neutral NAM branch; WET is post-cabinet.

The diagram describes NORMAL. In PLUS the NAM / IR chain runs at NAM 1's rack position and NAM 2 is skipped; the rack IR stays separate. PRE remains the independent neutral NORMAL NAM 2 branch, not a tap of the PLUS chain.

BLOCK NAM 2 and IR are **anchors**: they cannot be dragged, and no block can be moved so that IR ends up before BLOCK NAM 2. Everything else is free.

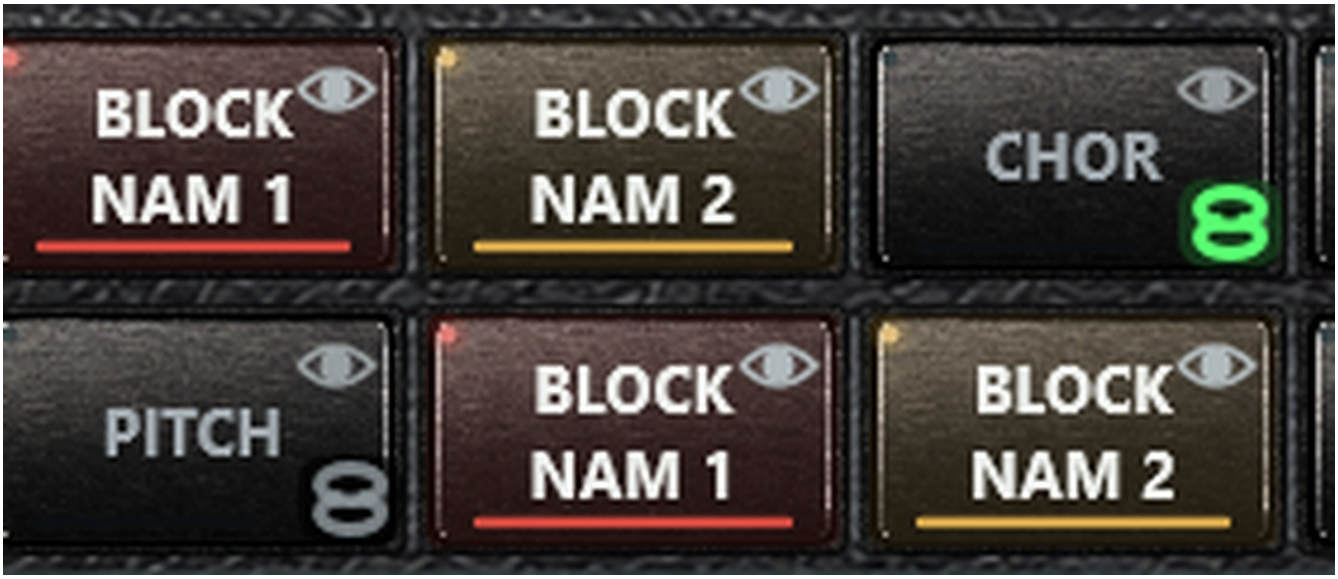
7. The rack: blocks, order and the eye



The two rack rows in STEREO: BLOCK NAM 1 (red), BLOCK NAM 2 (gold), an enabled CHOR, and the chain icons that link L and R

In NORMAL, eleven blocks per path, shown left to right in processing order. The default order is COMP · OCT · PITCH · BLOCK NAM 1 · BLOCK NAM 2 · CHOR · FLANG · PHASE · IR · DELAY · REVERB.

Reading a block



The parts of a block: name, eye, LED, colour bar and link icon

Part	Meaning
Body colour	Lit and tinted = enabled. Flat grey = bypassed
Small LED, top-left	Repeats the enabled state
Colour bar, bottom	The block's identity colour, lit while enabled
Eye, top-right	Chooses which block the shared controls and browsers describe. A viewed block has a green eye
Chain, bottom-right	Only in STEREO, only on effects: the L/R link. See chapter 13

What clicking does

Action	Result
Single click on the body	Enables or bypasses the block

Action	Result
Double-click on the body	Effects: opens the editor. BLOCK NAM 1 / BLOCK NAM 2: opens the model chooser. IR: opens the IR chooser
Click on the eye	Makes that block the one the shared controls edit. A bypassed block cannot be viewed
Drag horizontally	Moves the block; the target position is outlined in white
Drop a file on a block	Loads it: .nam on a NAM block, .wav on the IR block; a folder loads its first valid file
Hover BLOCK NAM 1 / BLOCK NAM 2	Shows the capture information card

A NAM or IR block can only be enabled once it holds a valid file. Clicking an empty one while its file is still loading means "switch on when ready", not "invert".

8. BLOCK NAM 1 and BLOCK NAM 2

Both blocks run any compatible .nam capture; the player does not assume what was captured. BLOCK NAM 1 sits earlier in the chain and is the natural place for a pedal or preamp capture, BLOCK NAM 2 for an amplifier - but nothing stops you using them the other way round.



The two browsers: NAM models on the left with its four selectors, IR files on the right, IR VOL between them

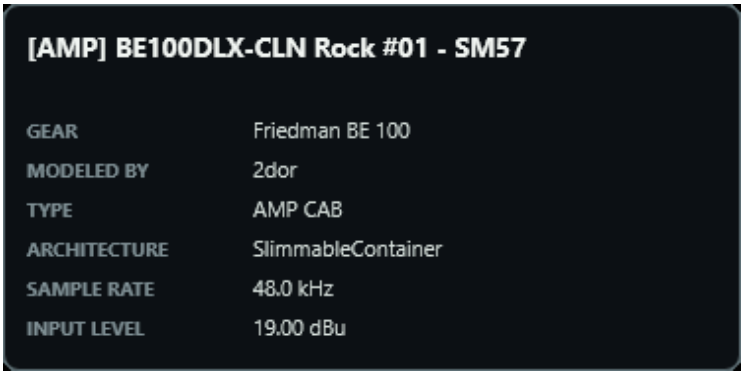
Loading a capture

Route	How
BROWSE LOCAL	Pick one .nam, or a folder. A folder adds every capture it contains to the list and selects the first. Tick **DEEP SEARCH** in the file dialog to include subfolders
The list	Click any row to load it; the ▲ / ▼ buttons step through the list and grey out at the ends
TONE3000	Opens the online library inside the player - see chapter 16
Drag and drop	Drop a .nam or a folder on the list or directly on the block
Double-click the block	Opens the file chooser for that block

Each block remembers its own folder, its own deep-search setting, its own list and its own selection, per path. Selecting a capture for BLOCK NAM 2 also resets that path's controls to safe defaults and switches the block on; the cabinet selection is left alone.

Architecture is automatic. The player identifies NAM A1, A2 Standard and A2 Nano and dispatches A2 models through NAM Core's fast path. There is no manual switch. If a file is not a valid single-input/single-output NAM, the load is refused and the block stays as it was.

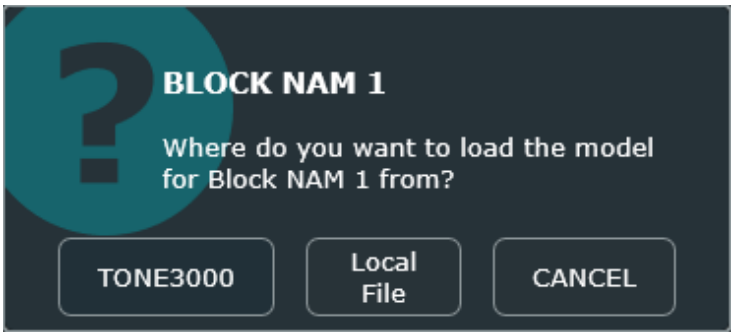
The capture information card. Hovering a NAM block or any row of the NAM list shows the metadata written into the .nam header - title, gear make and model, who modelled it, gear type, architecture, sample rate and input reference level - plus artwork when available.



The capture information card

Artwork is any image sitting beside the capture with the same base name (My Capture.nam → My Capture.png, .jpg, .jpeg or .webp). TONE3000 downloads save it automatically; for your own captures, drop an image next to the file. Only the header of the file is read, so hovering a long list costs nothing.

Loading BLOCK NAM 1. Choose its NAM selector to target the browser, or double-click its rack block to choose the capture source:

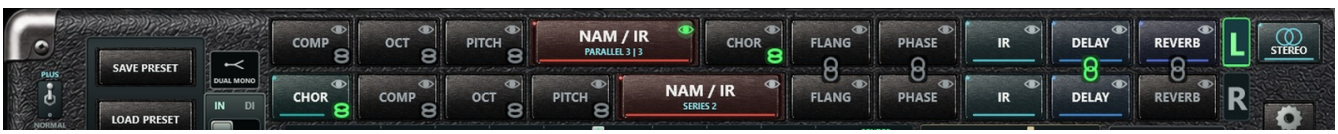


Choosing the source for BLOCK NAM 1

NORMAL / PLUS: NAM and IR chains

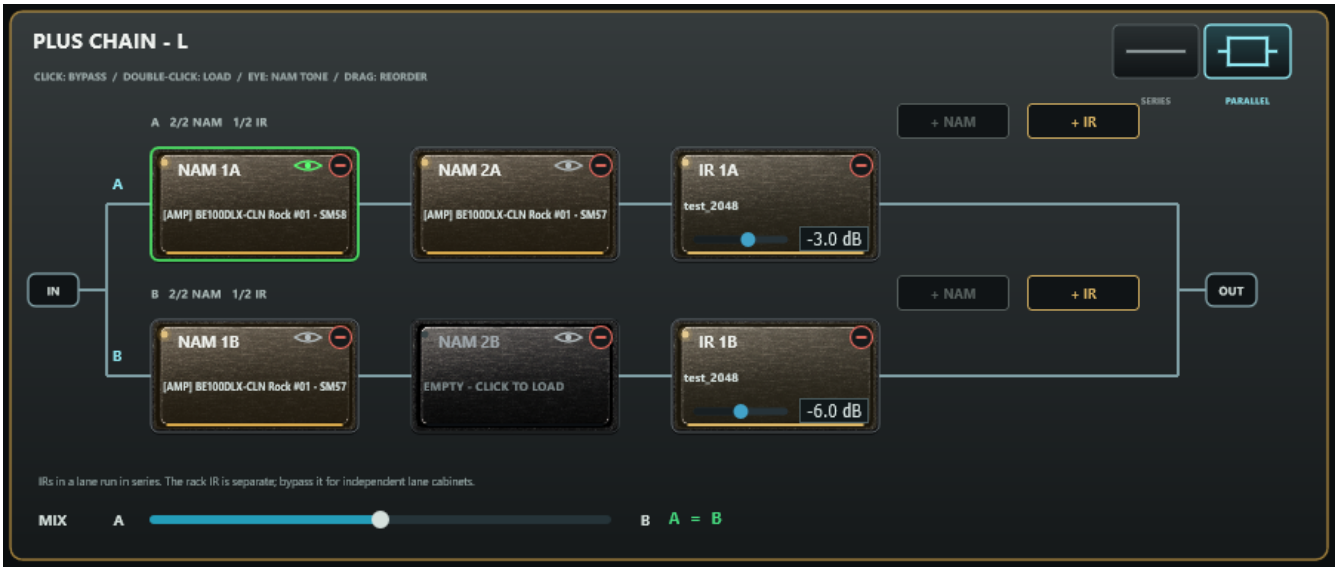
The lever switch on the left end of the rack chooses between two ways of using NAM captures:

Position	What the rack shows
NORMAL (lever down)	BLOCK NAM 1 and BLOCK NAM 2, exactly as described above
PLUS (lever up)	One large **NAM / IR** block, twice as wide, at BLOCK NAM 1's position in the chain



PLUS: one double-width NAM / IR block per path, with its chain topology under the name

The NAM / IR block of PLUS runs a **chain** of NAM and cabinet IR blocks. It switches on and off with a click and has the eye like any other NAM block; the line under its name shows the chain, for example SERIES 2 or PARALLEL 2 | 1. **Double-click it** to open the PLUS CHAIN window:



The PLUS CHAIN window in parallel: lane A above, lane B below, the + buttons and the A / B MIX

Element	Use
Single-line / parallel-lines switch	**SERIES**: one lane from IN to OUT. **PARALLEL**: two lanes, A above and B below, fed by the same input and mixed back together
NAM 1A / 2A and IR 1A / 2A (likewise B)	Up to two NAMs and two IRs per lane. Each lane retains one NAM placeholder. Blocks run in the displayed order
Click on a block	On / off (an empty block opens the load menu instead)
Double-click or right-click	NAM: load a local capture or use TONE3000. IR: load a local WAV, AIFF or FLAC. Remove block is available when allowed
Drop a file	Drop .nam on a NAM card or + NAM; drop WAV, AIFF or FLAC on an IR card or + IR
Eye (NAM cards only)	Select the NAM block edited by BASS, MID, TREBLE, PRESENCE and loaded from the NAM MODELS list or TONE3000
- (top right corner)	Remove the block, after confirmation. The last NAM of a lane cannot be removed; all IRs can be removed
+ NAM / + IR	Add the chosen type; each button becomes unavailable at its two-block limit. New NAMs are inserted before IRs
IR level	-24 to +12 dB, initially 0 dB; double-click to reset. WAV, AIFF and FLAC, mono or stereo averaged to mono, up to 8192 source samples, resampled without normalization
Drag a block	Reorder it, also into the other lane (if its two-NAM / two-IR limits allow it)
MIX A - B (PARALLEL only)	Linear crossfade: centre gives 50% of each lane; the ends give 100% of A or B. Unlike PAN, centre is not full level on both lanes

Each NAM block keeps its own capture, on/off switch and tone controls, just like BLOCK NAM 1 and BLOCK NAM 2. In PLUS the NAM selectors above the tone plate and over the list become one per path, captioned with the block being edited (NAM 2A), and the list shows the NAM models of that block.

The first time PLUS is switched on, loaded NORMAL NAM captures are copied into lane A in rack order, with their bypass and tone settings. An empty NAM placeholder remains if neither capture is loaded. The general rack IR is not copied into a lane. NORMAL and PLUS keep separate settings: going back to NORMAL finds BLOCK NAM 1 and BLOCK NAM 2 exactly as they were. The switch, the chains and their captures are saved with the session, in presets (the .bnpp bundle embeds every chain capture and IR) and, in the plug-in, with the DAW project. Every change of the chain is applied behind a short fade, with a short transition to suppress clicks. With INPUT CALI on, PLUS calibrates to the first active NAM of lane A; standalone DI playback bypasses interface calibration.

IRs within a lane run in series. To mix two cabinets, put one in each parallel lane and use A/B MIX. The rack IR remains a separate shared block: bypass it when using independent lane cabinets to avoid filtering the signal twice. IR files load in the background, and their level, order and bypass state are saved with the chain. Legacy chains with

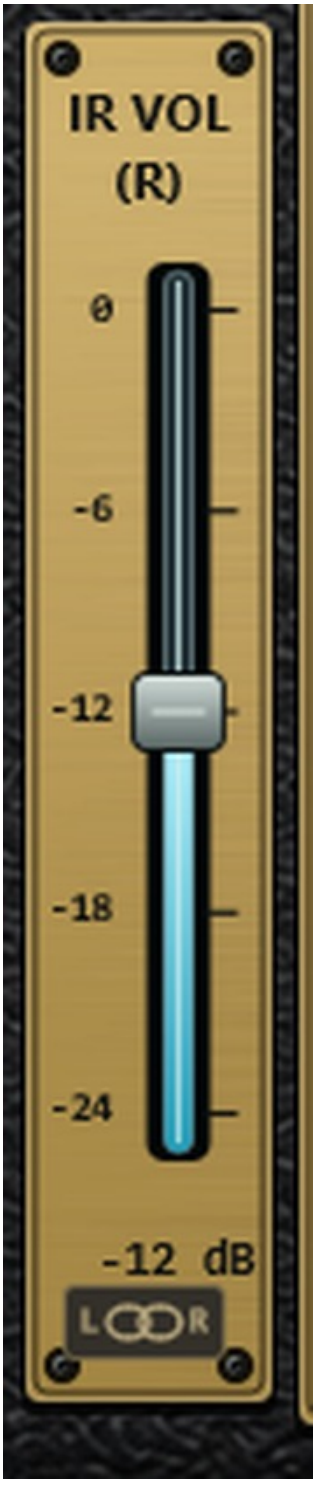
more than two NAMs per lane restore the first two and show a notice; keep the original preset if you need its older configuration.

9. Cabinet impulse responses

BROWSE IR loads a .wav response, or a folder: the player keeps the short WAV files that are usable as impulse responses, lists them and selects the first. If nothing suitable is found it reports **IR not found**.

- Mono and stereo responses are accepted; a stereo file is summed to mono.
- Up to 8192 samples are used, resampled to the current device rate.
- **The original gain is preserved - IRs are never normalized.**
- Click the selected row a second time to deselect it: the response is cleared and the IR block is bypassed.
- **DELETE** removes the selected response from the IR folder, after asking.
- Loading or changing a response crossfades, so it never clicks.

Two controls shape the cabinet:



IR VOL, with its own scale and the (L)/(R) caption in STEREO

- **IR BLEND** mixes the signal without cabinet convolution against the convolved signal, 0 - 100 %.
- **IR VOL** (the vertical fader) sets the level of the cabinet branch alone, **-24 dB to 0 dB**, default **-12 dB** to leave headroom for hot captures. The current value is printed under the scale.

10. Amplifier controls, knob by knob



The main control row and the NAM/IR selectors above it

Operating a knob: drag vertically or horizontally; the exact value is printed underneath. **Double-click restores the default** - 12 o'clock for everything except GATE, which returns to fully left (OFF).

Control	Range	Default	Notes
INPUT GAIN	-12 … +12 dB	0.0 dB	Level into the NAM chain; Input Cali is added on physical IN; bypassed for DI files
GATE	OFF … -80 … 0 dB threshold	OFF	At the far left it is a true bypass. Attack 1.5 ms, hold 35 ms, release 90 ms, 3 dB hysteresis
BASS	±12 dB @ 70 Hz	0.0 dB	Peaking, Q 0.72
MID	±12 dB @ 750 Hz	0.0 dB	Peaking, Q 0.72
TREBLE	±12 dB @ 4 kHz	0.0 dB	Peaking, Q 0.72
PRESENCE	±12 dB @ 6 kHz	0.0 dB	Peaking, Q 0.72
MASTER VOL	-12 … +12 dB	0.0 dB	After the chain, before Output Gain
IR BLEND	0 … 100 %	50 %	Dry against cabinet
IR VOL	-24 … 0 dB	-12 dB	Cabinet branch only
OUTPUT GAIN	MUTE, then -60 … +12 dB	0.0 dB	Final MAIN level of the selected path; centre is unity
POWER	On / off	On	Mutes the processed path; the switch glows red while on
INPUT CALI	On / off	Off	See chapter 15

At the far-left stop, **OUTPUT GAIN** shows **MUTE** and the selected MAIN path becomes completely silent after a short 10 ms ramp. Above that stop the lower half rises from -60 dB to 0 dB at the centre; the upper half reaches +12 dB. L and R can be muted independently or together with their link. The standalone's separate PRE, DIRECT and WET hardware sends are reference taps and are not controlled by MAIN OUTPUT GAIN. If CAB POS places the other, unmuted path in the same physical speaker, that speaker may still sound: OUTPUT GAIN mutes the selected path, not a speaker. Link L/R and turn the knob to MUTE to silence both MAIN paths.

The tone stack follows the selected NAM block. With a **NAM 1** selector active, BASS / MID / TREBLE / PRESENCE drive BLOCK NAM 1's own four filters; with a **NAM 2** selector active they drive the main tone stack. Each set keeps its own values, and switching selectors recalls them. In the plug-in and in STEREO the plate prints which one you are editing.

In STEREO, INPUT GAIN and OUTPUT GAIN show (L) or (R) beside their labels to identify the currently edited path. Every path-dependent control has its own small **L-chain-R** switch, except TUNER. A dark switch keeps L and R independent. Select L or R, set its value, then light that control's link: its current value is copied to the other path. Later editor changes to that one control update both paths; turning the link off leaves their current values in place for separate editing. Links are per parameter, not a master stereo link: INPUT GAIN can be linked while OUTPUT GAIN remains independent. The POWER link beside its red switch makes that switch turn both paths on or off together; it does not link gain, tone or the audio signals. INPUT CALI has its own link beside its indicator. Tone links follow the

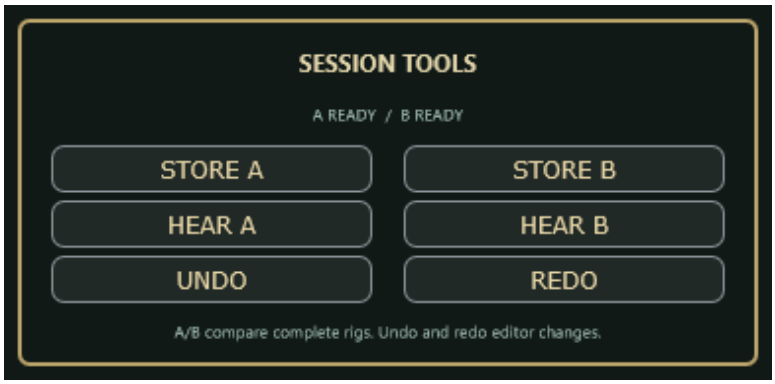
selected NAM block on each path, including the viewed NAM slot in PLUS. TUNER monitors only the selected path, so it has no link. Links are saved in the standalone session and in .bnpp presets. The separate effect-link icons on the rack continue to link whole effects.

RESET NAM, beside the NAM MODELS header, asks for confirmation. **NAM ONLY** returns the selected NAM tone and its path's MASTER VOL / OUTPUT GAIN to their defaults. **ENTIRE PATH** resets all front-panel controls on the selected L or R path; in PLUS it also resets that path's NAM tones and slot levels. Neither choice deletes NAM or IR files or changes the effect order. Controls whose L/R link is lit also update the other path. The button is unavailable while an IR slot, rather than a NAM, is selected in PLUS.

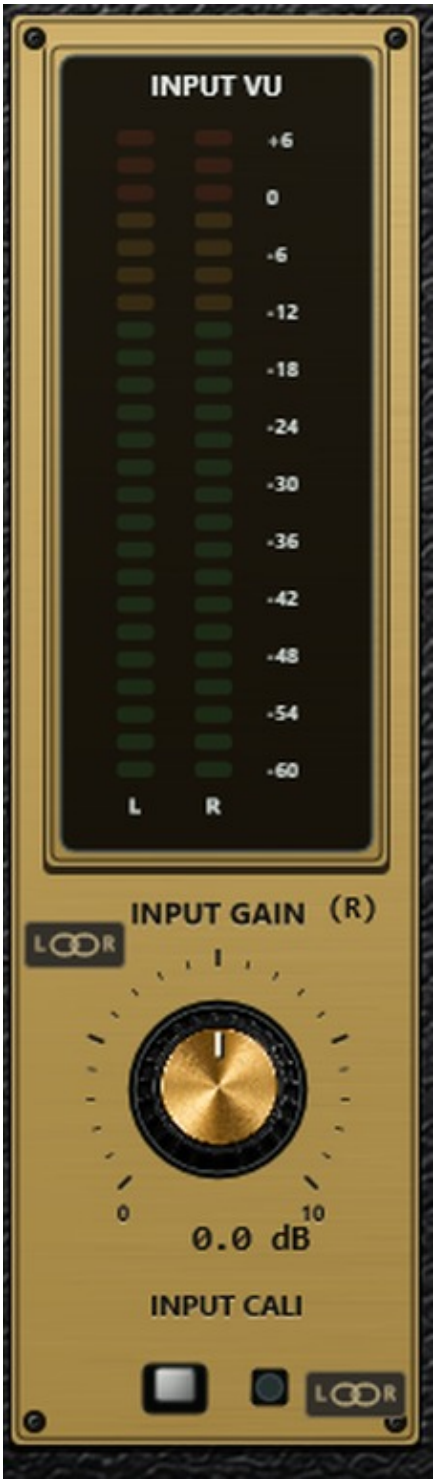
Session tools: A/B and history

SESSION TOOLS is the small gold-edged button to the right of IR FILES. It opens a closable hardware-style window and leaves the main controls clear. **STORE A** and **STORE B** capture two complete rigs: NAM and IR choices, effects, levels, routing, stereo positions and NORMAL / PLUS chains. **HEAR A** and **HEAR B** recall either rig, so you can compare them using the same guitar part or DI playback. The DI source, files, transport and loop are not part of these rig snapshots. Output level is recalled exactly; there is no automatic loudness matching or normalization.

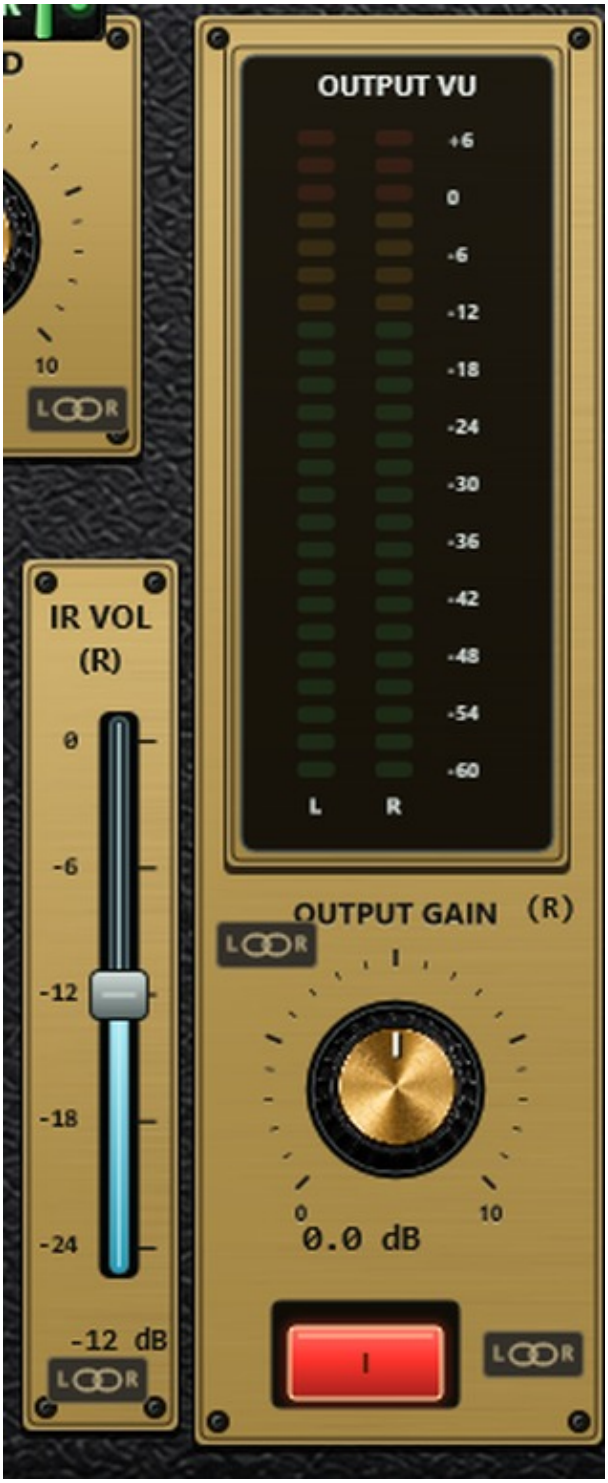
UNDO and **REDO** walk through up to 40 recent rig changes, including control edits, links, block order and loaded files. A continuous knob drag is grouped after it settles. Switching A/B also enters the history, so UNDO can return to the sound heard before the switch. These working states last only while the application is open; they are not embedded in .bnpp presets. File changes use the existing smooth output transition.



Session tools: store and hear two rigs, undo and redo



Input column: Input VU, INPUT GAIN and INPUT CALI with its LED



Output column: Output VU, OUTPUT GAIN and POWER

11. The eight effects in full

The eight effects offer **three algorithms** and **six main parameters** each. PITCH also includes the HARMONIZER controls described below. Double-click a block to open its editor; every knob shows a real unit, and double-clicking a knob restores its default.



The effect editor, here for DELAY

The **ACTIVE** switch at the top right of the editor is the same bypass as clicking the block. **TYPE** chooses the algorithm. Parameter, algorithm and bypass changes are smoothed, delay-family reads are interpolated and feedback is bounded, so nothing clicks or runs away.

COMP - compressor

Algorithms: **Studio VCA**, **Optical**, **FET Punch**.

Parameter	Range	Default
Threshold	-55 … -2 dB	-31.2 dB
Ratio	1 … 20 :1	7.7:1
Attack	1 … 100 ms	15.9 ms
Release	20 … 600 ms	223 ms
Makeup	-12 … +12 dB	0.0 dB
Mix	0 … 100 %	100 %

DELAY

Algorithms: **Digital Studio**, **Tape Echo**, **Analog BBD**.

Parameter	Range	Default
Time	20 … 1200 ms	398 ms
Feedback	0 … 92 %	32 %
Mix	0 … 100 %	28 %
Tone	0 … 100 %	65 %
Mod	0 … 100 %	8 %
Level	-12 … +12 dB	0.0 dB

CHOR - chorus

Algorithms: **Studio**, **Ensemble**, **Tri-Chorus**.

Parameter	Range	Default
Rate	0.05 … 5 Hz	1.24 Hz
Depth	0.5 … 20 ms	10.3 ms
Mix	0 … 100 %	35 %
Delay	4 … 30 ms	11.8 ms
Feedback	-65 … +65 %	0 %
Level	-12 … +12 dB	0.0 dB

FLANG - flanger

Algorithms: **Analog**, **Through-Zero**, **Jet**.

Parameter	Range	Default
Rate	0.03 … 2.5 Hz	0.57 Hz
Depth	0.1 … 9 ms	5.0 ms
Mix	0 … 100 %	35 %
Feedback	-85 … +85 %	20 %
Manual	0.2 … 5 ms	1.4 ms
Level	-12 … +12 dB	0.0 dB

PHASE - phaser

Algorithms: **4 Stage**, **8 Stage**, **12 Stage**.

Parameter	Range	Default
Rate	0.03 … 4 Hz	0.82 Hz
Depth	0 … 100 %	70 %
Mix	0 … 100 %	40 %
Feedback	-75 … +75 %	12 %

Parameter	Range	Default
Centre	180 … 2200 Hz	887 Hz
Level	-12 … +12 dB	0.0 dB

REVERB

Algorithms: **Studio Room**, **Plate**, **Concert Hall**.

Parameter	Range	Default
Size	0 … 100 %	55 %
Damping	0 … 100 %	50 %
Mix	0 … 100 %	25 %
Width	0 … 100 %	80 %
Freeze	OFF / ON	OFF
Level	-12 … +12 dB	0.0 dB

OCT - octaver

Algorithms: **Poly Clean**, **Classic Mono**, **Organ**.

Parameter	Range	Default
Oct Down	0 … 100 %	45 %
Oct Up	0 … 100 %	0 %
Dry	0 … 100 %	80 %
Tone	0 … 100 %	50 %
Tracking	0 … 100 %	65 %
Level	-12 … +12 dB	0.0 dB

PITCH - pitch shifter

Algorithms: **Studio**, **Low Latency**, **Vintage**.

Parameter	Range	Default
Semitones	-12 … +12 st	0.0 st
Mix	0 … 100 %	100 %
Window	20 … 120 ms	65 ms
Feedback	0 … 55 %	0 %
Fine	-100 … +100 ct	0 ct
Level	-12 … +12 dB	0.0 dB

HARMONIZER (scale-aware harmony)



The PITCH window with the HARMONIZER strip: switch, INTERVAL, KEY, SCALE, TUNING and the live display

The strip at the bottom of the PITCH window turns the block into an intelligent harmonizer. With **HARMONIZER** on, every note you play gets a second voice a scale step away, using the selected or detected scale:

INTERVAL	Harmony voice
OCTAVE UP / OCTAVE DOWN	Always an octave; needs no key, sounds from the first note and also works on chords
THIRD UP / THIRD DOWN	The third of the scale: **major or minor depending on the note**. In C major, C gets E (major third) and D gets F (minor third); in C minor, C gets Eb
FIFTH UP / FIFTH DOWN	The fifth of the scale: perfect, or diminished on the seventh degree (B -> F in C major)

KEY and SCALE. With KEY on **AUTO** the key is learned from the notes you play:

- It works on the seven notes in use, which is what decides the harmony. A key is told apart from its parallel (C major / C minor) by the notes actually played (E or Eb, A or Ab, B or Bb); a key, its relative and its modes (C major, A minor, D dorian...) share their notes and therefore their harmony.
- The tonic and the mode are named from where the playing dwells and, above all, where phrases come to rest: the display can read A MINOR, D DORIAN, G MIXOLYDIAN... Pentatonic playing is read as the natural minor / major until other notes say otherwise. **Harmonic minor** is recognised when the raised seventh is used throughout (G# and never G in A minor): the dominant then gets its major third.
- Two memories run side by side: a long one keeps the key steady and a short one follows a real key change within a few seconds. The detector resists short passing notes; fix KEY manually if automatic detection does not match your phrase.
- Until enough notes have been heard the display shows **LISTENING...** and thirds and fifths stay silent, until detection has enough confidence. The bar under the key shows how sure the detection is.

Choose a tonic in **KEY** to fix the key yourself; **SCALE** then offers major, minor, the five other modes, harmonic minor and melodic minor.

TUNING. PURE tunes each interval to the played note with simple ratios (5:4 and 6:5 thirds, 3:2 fifths): the two voices lock together without beating, which is what keeps a harmony clean through an overdriven amp. **TEMPERED** uses the piano's equal semitones, to match keyboards exactly.

The voice. The harmony is made by a shifter of its own whose splices are synchronised to the period of the note being played, so it sounds like a second guitar rather than an effect. It reads the guitar a few milliseconds late - the natural delay of a second player - and uses that time to know each new note before it sounds: a note never comes out with the interval of the previous one. Notes are recognised from low E (and drop tunings) up to the 24th fret, within a few cents, in about 25 ms (45 ms on the lowest strings). Bends and vibrato are followed continuously: in a bend from C to D in C major the harmony slides from E to F. Notes outside the key (a blue note, the G# of A minor) borrow the scale degree that gives a real third or fifth: G# gets B, Bb in C major gets D.

Knobs in HARMONIZER mode: **MIX** balances the harmony against the note you play, which always stays (100 % = both at the same level, 0 % = no harmony); **FINE** detunes the harmony slightly for a wider sound; **LEVEL** works as usual. SEMITONES, WINDOW and FEEDBACK rest. TYPE chooses the character of the voice: **Studio**, **Low Latency** (a shorter delay; very fast phrases may lose a little accuracy on low notes) and **Vintage** (darker). The rack block reads **HARMONY** while the harmonizer is on.

Play single notes: thirds and fifths follow melodies, riffs and solos. For thirds and fifths, use single-note playing. Polyphonic or unclear input can make pitch detection unreliable; the detector may suppress the harmony. Detection uses the clean guitar signal, before gain, NAM and effects. The HARMONIZER settings are saved with the PITCH block and follow an L / R link.

The dry guitar remains immediate while the added harmony voice uses a short, variable pitch-analysis delay. **Low Latency** shortens that voice delay; it does not change the timing of the original guitar.

12. MONO and STEREO: two complete rigs



STEREO with SPLIT L/R: two rack rows, four NAM selectors, two IR selectors and stereo meters

MONO / STEREO (top right of the rack) switches between one and two audible paths.



The MONO / STEREO switch and the settings gear

- **MONO**: the interface input is summed into the left chain, and its result feeds both channels of every routed output pair. Right-hand selectors, the second rack row and the input-mode switch are hidden.
- **STEREO**: both chains run. Two rack rows appear, the meters split into L and R, and the L / R tabs beside the racks choose which path the shared controls edit.

DUAL MONO / SPLIT L/R (beside the preset bay, STEREO only):

Position	Input handling
DUAL MONO	The summed interface input feeds **both** chains - one guitar through two independent rigs
SPLIT L/R	Input channel 1 feeds the left chain, input channel 2 feeds the right chain

This setting belongs to the application state, not to presets. The table describes physical IN. In DI mode MONO and DUAL MONO use the left file, while SPLIT L/R uses one file per path (chapter 6).

PAN (the bar between the two rack rows and the tuner, STEREO only): a horizontal fader that balances the level of the two paths.

Position	Result
Centre (CENTER, centre notch, green mark)	L and R play at their full level, exactly as without PAN
Towards L	The right path fades out progressively; at the left stop (L 100 %) only L is heard

Position	Result
Towards R	The left path fades out progressively; at the right stop (R 100 %) only R is heard

The L and R letters at the ends are lamps: the one on the side being faded dims, and the rail lights up from the centre towards the favoured side. Neither side ever gets louder than it is at the centre. Double-click returns the fader to the centre; the mouse wheel moves it in fine steps.

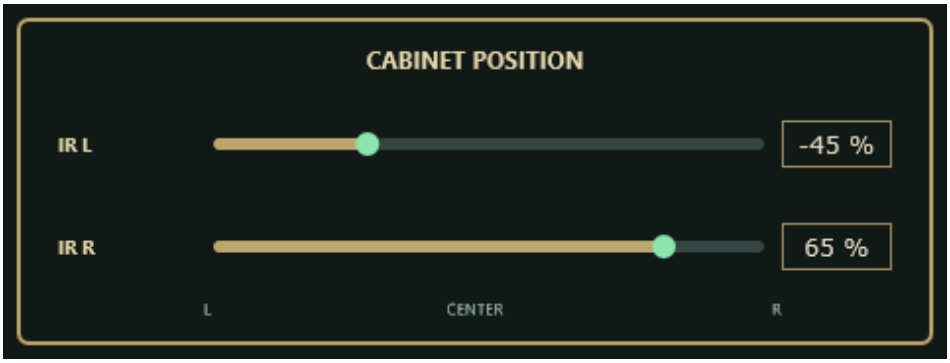
In the standalone PAN acts on the main and WET outputs (the PRE and DI taps stay untouched) and is saved with the application state and in presets.

OUT is a shared post-chain output trim (-60 to +12 dB, 0 dB by default). It changes both processed channels together without changing their relative balance. Double-click the strip to return to 0 dB. In the standalone it affects MAIN and WET, not the PRE and DI reference taps. In the plug-in it acts on the final stereo output.

Cabinet position

CAB POS opens the stereo cabinet-position window. **IR L** positions the entire left processed cabinet path in the final stereo field; **IR R** does the same for the right path. Initially L is hard left and R hard right. For example, move IR L to CENTER to hear the left cabinet from both speakers while IR R stays on the right. Set both to CENTER to hear both cabinets from both speakers. These sliders do not load, select or blend IR files. In PLUS, two IRs in one lane are in series and CAB POS positions that lane's combined result.

PAN subsequently balances the complete L and R paths; it cannot move only one cabinet. **OUT** changes their final level together; **OUTPUT GAIN** belongs to the selected path and may be linked L/R. CAB POS moves are smoothed to avoid clicks. Its two positions and OUT are saved with the standalone session and .bnpp presets.



Cabinet position controls for the left and right processed paths

The selectors.



NAM 1 L · NAM 2 L · NAM 1 R · NAM 2 R and IR L · IR R

Selector	Chooses
NAM 1 L NAM 2 L NAM 1 R NAM 2 R	The NAM block the tone controls and the NAM browser act on
IR L IR R	The cabinet that IR BLEND and IR VOL act on
L / R tabs	The path every other shared control acts on

Which control follows what

Control	Follows
INPUT GAIN, GATE, MASTER VOL, OUTPUT GAIN, POWER, INPUT CALI	the selected **path**; each has its own optional L/R link
BASS, MID, TREBLE, PRESENCE	the selected **NAM block** of that path (NAM 1 has its own set)
IR BLEND, IR VOL	the selected **cabinet**
TUNER	the selected path only; no L/R link or two-note mixing
PAN, OUT	both processed paths together; neither follows the path tab
CAB POS IR L / IR R	independent stereo placement of each complete cabinet path
Skin, Audio Setup, routing, updates	the whole application

In MONO only the two left selectors are shown, and the IR VOL legend drops its (L) / (R) suffix.

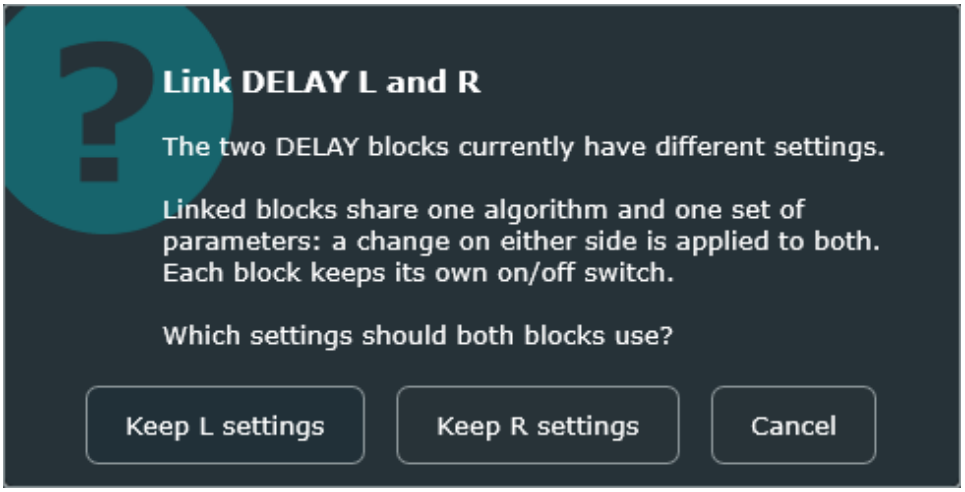
13. Linking effects between L and R

In STEREO you can lock an effect of the left chain to the same effect of the right chain, so you only have to dial it once.



Linked pairs are green; unlinked pairs are grey. A pair that no longer shares a column shows its icon in the corner of both blocks

- **Where the icon is.** When both blocks are in the same column, one chain icon sits on the seam between the two rack rows and joins them. If you move one of them so the pair no longer shares a column, the icon moves to the **lower-right corner of each block** - it still works exactly the same.
- **Colour.** Green = linked. Grey = not linked.
- **Click to link or unlink.**
- **What is shared:** the algorithm and the six parameters. Editing either block changes both.
- **What is not shared:** the on/off switch and the position in the chain. The two blocks can be both on, both off, or one of each - linking never touches them.
- **Linking two blocks that differ** asks which side to keep:



Choosing which side's settings to keep

- **Unlinking** leaves both blocks exactly as they sound at that moment; from then on they are edited separately.
- **BLOCK NAM 1, BLOCK NAM 2 and IR cannot be linked.**
- Links are saved with the session and inside two-path .bnpp presets.

14. Tuner



The tuner row: the TUNER button, LED display and CHROMATIC indicator

Press **TUNER**; the lettering lights green and the button looks pressed. The display shows the detected note with a left/right LED indication for flat, centred or sharp.

- The tuner listens to the **untouched DI**, before the gate, the NAM blocks and the effects, and is not part of the draggable order.
- Range roughly **65 - 700 Hz**, at every sample rate from 44.1 to 192 kHz.
- Within **±5 cents** the display reads centred and turns green; beyond about 40 cents it turns red.
- **CHROMATIC** reports the actual detected note regardless of guitar tuning. It does not assign a string or target note: compare the display with the note you intend to tune that string to.
- In STEREO, choose the L or R path to tune. The tuner does not combine two inputs; its on/off state remains independent for each path.
- Play one isolated string at a useful level and let the previous note decay.

15. Input Cali (automatic input calibration)

NAM captures are trained at a known input level. Input Cali matches your interface to that level so a capture sounds the way it was captured.

With Input Cali on, the player computes:

calibration gain (dB) = interface input reference (dBu) - NAM input reference (dBu)

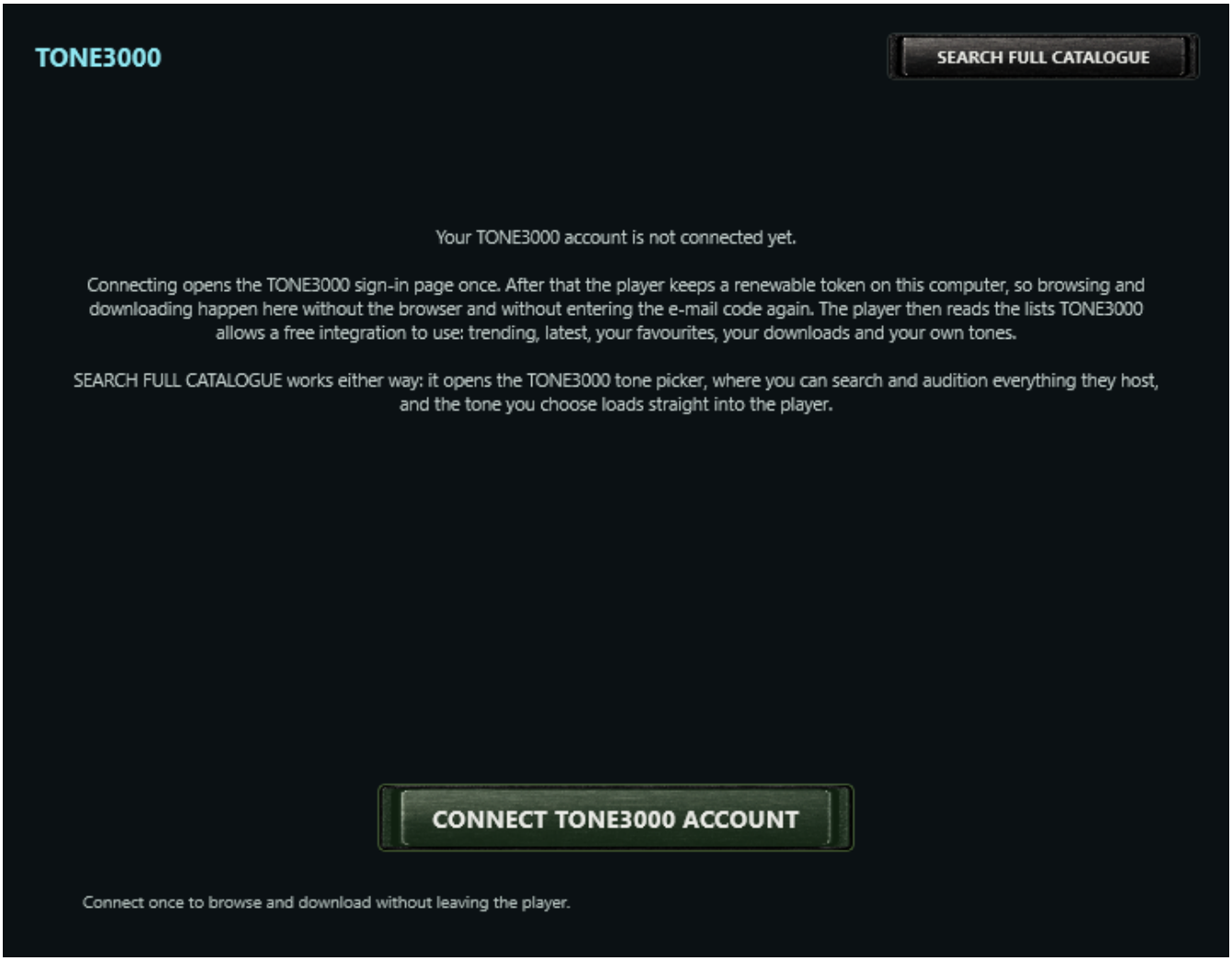
- The interface value is the slider in Audio Setup.
- The NAM value comes from the capture's input_level_dbu metadata.
- When a capture has no metadata, the standard NAM reference of **+12 dBu** is assumed.

- The result is clamped to **-24 … +24 dB** and applied before the NAM blocks.
- The small LED beside INPUT CALI is green while it is active. The DI source bypasses interface calibration; returning to IN restores its setting.
- While a replacement capture is still loading, the gain of the running capture is held - it never jumps to 0 dB mid-note.

Input Cali changes gain only. It never rewrites or normalizes a model. If your interface has several input modes (instrument / line / pad), enter the dBu value of the mode you are actually using, as specified by the interface manufacturer.

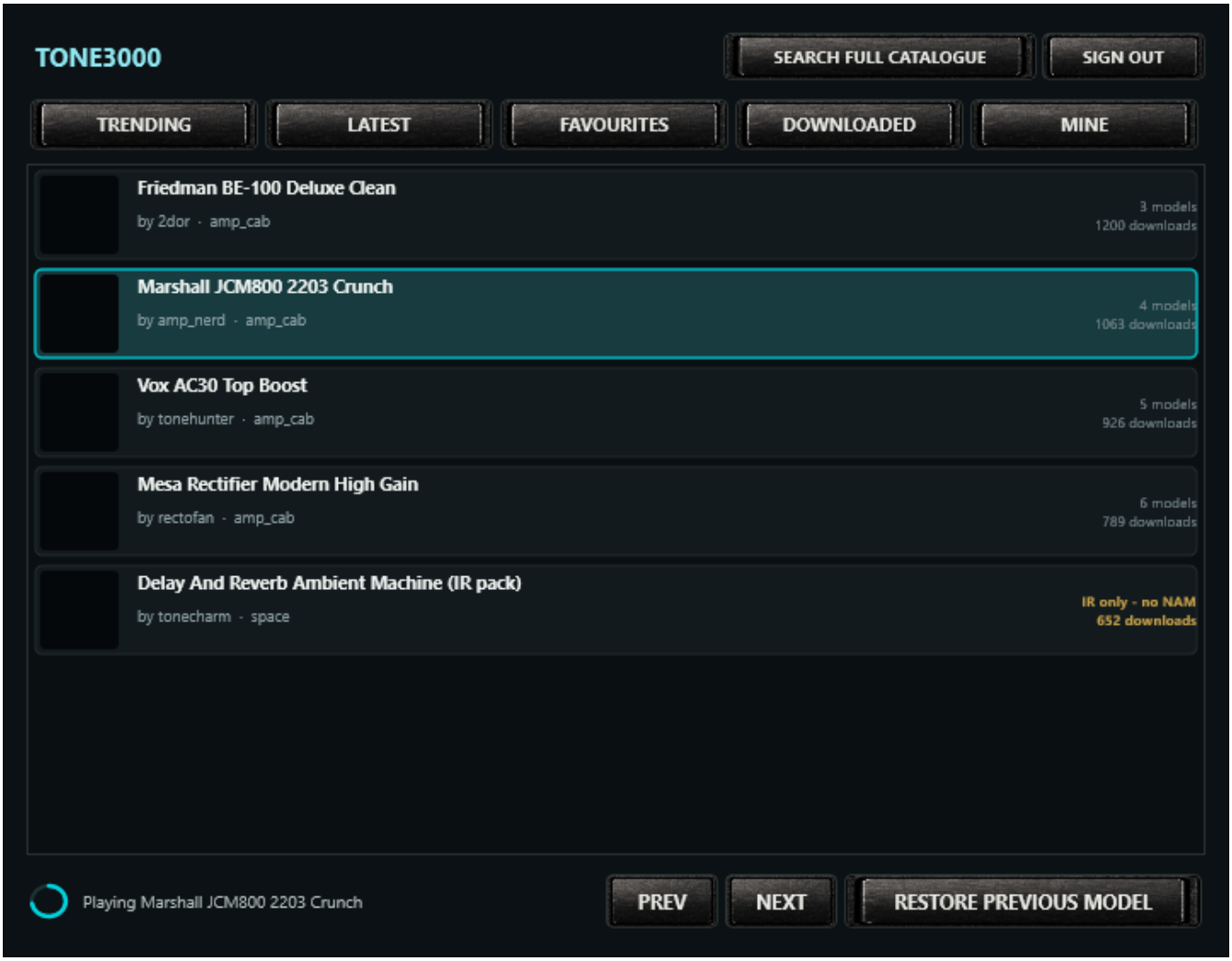
16. TONE3000 inside the player

TONE3000 (green button above the NAM list) opens the online capture library in a window of the player.



Before connecting an account

Connecting once. The first use asks to connect your account. TONE3000 signs in with an e-mail link, so that opens their sign-in page once; the player then keeps the returned refresh token, encrypted with a key derived from this machine, in tone3000.session beside the application. Every later listing and download normally renews the token silently on that computer while the saved authorization remains valid. If it expires or is revoked, reconnect. **SIGN OUT** deletes the file.



The library: five lists, artwork, author, model count and paging

Browsing and auditioning.

Element	What it does
TRENDING · LATEST · FAVOURITES · DOWNLOADED · MINE	The five listings a free-tier integration may read
Selecting a row	**Loads the capture into the current NAM block immediately.** The first capture of the tone is fetched and plays as soon as it is on disk; the rest of the tone and its artwork keep downloading in the background
Selecting another row	Cancels whatever is still downloading and auditions the new one instead
A tone you already downloaded	Loads instantly from your downloads folder
Rows marked **IR only - no NAM**	Contain only impulse responses; there is nothing to load into a NAM block, so they are not auditioned
RESTORE PREVIOUS MODEL	Puts back the capture the block held **before this window was opened**. Disabled if the block was empty
Closing the window	Keeps the last capture you auditioned
PREV / NEXT	Paging
The turning indicator and the status line	Show progress: *Downloading tone (n / m)*, then *Playing <capture>*
SEARCH FULL CATALOGUE	Opens TONE3000's picker inside the player (EmbeddedBrowser) or in the system browser (ExternalBrowser). Select a tone to start its download; the embedded picker closes automatically
SIGN OUT	Forgets the account on this computer

Downloads are saved under your TONE3000 downloads folder, one folder per tone, with the tone artwork beside each capture so the information card can show it.

WebView2 (Windows only). The embedded sign-in and the catalogue picker use Microsoft WebView2, which ships with Windows 11 and with Edge on Windows 10. If it is missing or unavailable the player offers an official Microsoft download, your external browser, or cancelling - **nothing is ever installed automatically**, and no Microsoft installer is included in the package. Local audio and local files never need WebView2. macOS uses WKWebView; Linux falls back to the system browser. External and embedded browsers do not share cookies, so a session created in an older build may need one sign-in inside the player.

17. Portable .bnpp presets

SAVE PRESET writes one self-contained archive holding **both paths**:

- the BLOCK NAM 2 capture of each path;
- the BLOCK NAM 1 capture of each path, when loaded;
- the selected rack IR of each path, when loaded;
- NORMAL / PLUS mode and every PLUS NAM and IR file, lane order, bypass, tone, IR level and A/B mix;
- rack order, algorithms, all effect parameters and every bypass state;
- all amplifier knob, switch and tuner settings (not the DI player);
- the L/R effect links.

Embedded resources are verified with SHA-256 when loaded, extracted to the application's preset cache and restored. Audio device, interface reference and physical routing, DI files and DI transport are deliberately excluded so a preset can move between computers.

LOAD PRESET restores it. A preset written by the single-path player, or by NAM PLAYER DUAL, loads into the **currently selected path**. A preset written here includes a left-path compatibility representation. Older players cannot reproduce features they do not implement, including the current PLUS chain; use 1.8.5 to recall the full configuration.

> To save from NORMAL, load BLOCK NAM 2 on the left path. PLUS also permits saving with a loaded file in its left-path chain. DI files are never embedded.

18. What is remembered, and where

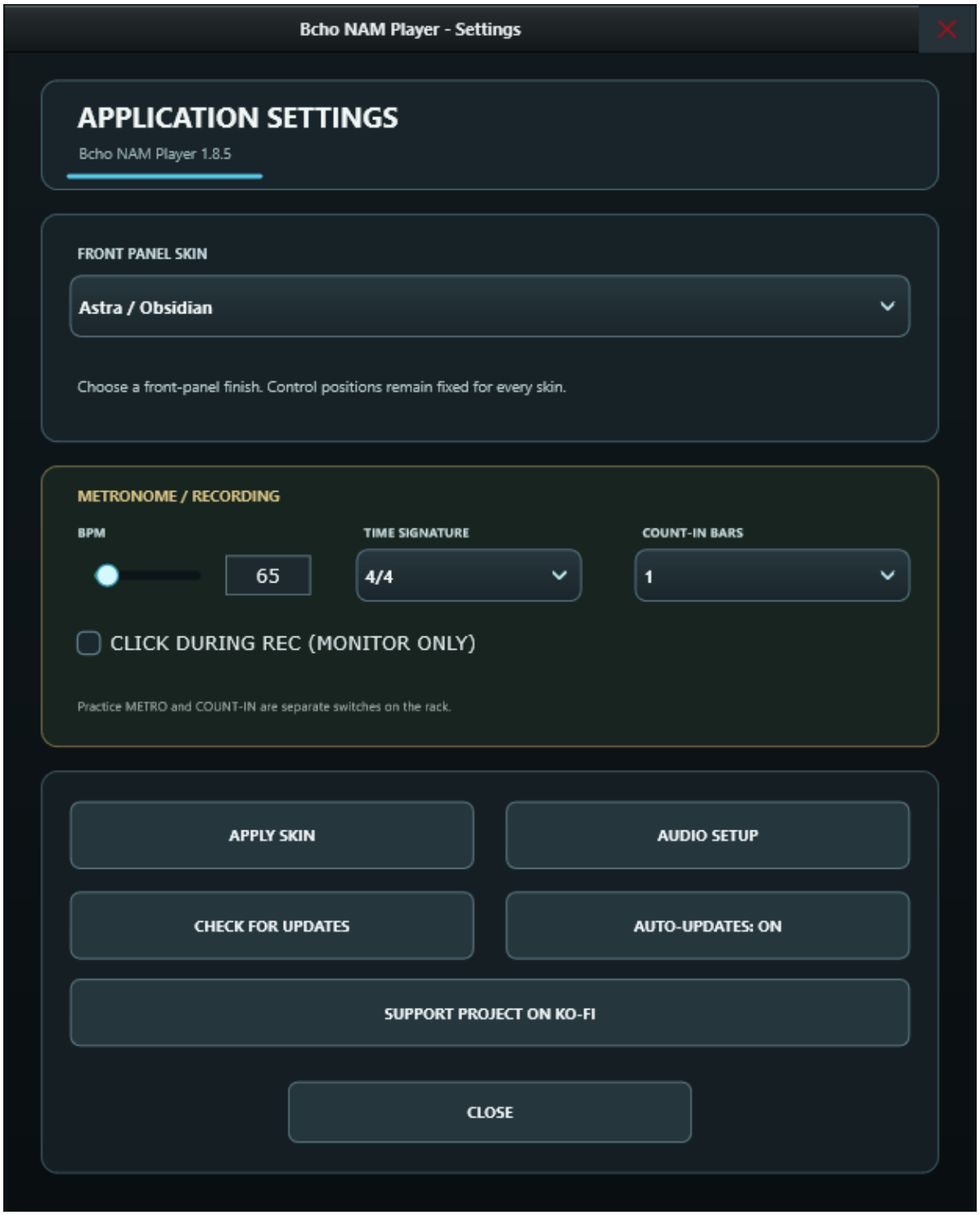
Session state is written on a normal exit and restored on the next launch: both paths, all controls, the selected captures and IRs, block order and bypass states, links, MONO/STEREO, DUAL MONO/SPLIT, skin, audio device, channels, rate, buffer, interface reference and the four routes. PLUS chains and DI file paths, levels, position and loop are also recalled. The source always starts on IN, with DI paused; switching back to DI never starts playback automatically.

File	Where	What it holds
NAM_A2_HEAD.xml	Beside the application when that folder is writable, otherwise the per-user data folder	The session
tone3000.session	Same folder	The encrypted TONE3000 refresh token
Presets / PresetCache	Same folder	Presets you save and the resources extracted from loaded ones
Per-user data folder	Windows %APPDATA%\Bcho\BchoNAMPlayerDual, macOS ~/Library/Application Support/Bcho/…, Linux ~/.config/Bcho/…	Used when the application folder is read-only (macOS bundles, AppImages)

NAM and IR files are referenced by path, not copied into the session: keep them where they were when you saved. Presets, by contrast, embed them.

19. Settings, skins, updates and support

Press the gear to the right of the rack.



Application settings

Control	What it does
FRONT PANEL SKIN	Previews a finish immediately

Control	What it does
APPLY SKIN	Makes the previewed finish permanent. Closing without applying restores the previous one
AUDIO SETUP	Opens Audio Setup
CHECK FOR UPDATES	Compares your version with the latest public release
AUTO-UPDATES: ON / OFF	Whether that check also runs at startup
SUPPORT PROJECT ON KO-FI	Opens ko-fi.com/bchosoft in your browser
CLOSE	Closes the window

The fifteen finishes: Astra / Obsidian · Tribal / Etched Titanium · Skulls / Bone & Carbon · Hippie / Sunset Paisley · Graffiti / Electric Ink · Purple Velvet / Amethyst · Stainless Steel / Precision · Ripped Black Denim / Roadworn · Blue Denim / Indigo · Spiderwebs / Black Widow · Classic Black / Levant Tolex · Classic Black / Purple Metal · Forest Green / Gunmetal · Midnight Blue / Silver · Graphite / Electric Blue. The purple finish retains the same black Levant tolex and replaces the gold hardware plates and metal knobs with reflective amethyst-purple metal. Forest Green has dark green textured tolex, gunmetal plates and blue-anodised knobs. Midnight Blue combines navy tolex, bright silver plates and silver knobs. Graphite uses charcoal tolex, dark brushed metal and smoked-chrome knobs with blue rings. Every finish uses identical geometry, so no control ever moves; only materials, knob designs and legend contrast change. The plug-in always uses Astra / Obsidian.



Classic Black / Purple Metal finish



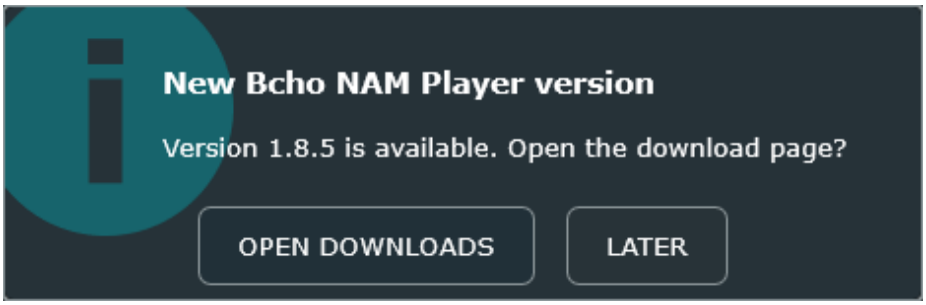
Forest Green / Gunmetal finish



Midnight Blue / Silver finish



Graphite / Electric Blue finish



The update notice

Update checking only reports availability and opens the download page if you ask it to. It never replaces files, models or presets.

20. Meters and animated cones

INPUT VU shows the incoming level and **OUTPUT VU** the final processed level, from +6 down to -60 dB. In STEREO each meter splits into an L and an R column.



The cabinet: both cones follow the real output level

Both cones react to the real final output RMS: louder playing, or a higher MASTER VOL / OUTPUT GAIN, gives a larger excursion; silence and POWER off return them smoothly to rest. Only the cone surfaces move - rings, screws, grille, cabinet and logo stay fixed. The animation runs at 60 frames per second on the interface thread and never touches, delays or feeds back into the audio.

21. Mouse and keyboard reference

Where	Action	Result
Any knob	Drag up/down or left/right	Change the value
Any knob	Double-click	Restore the default (GATE returns to OFF)
Rack block	Single click	Enable / bypass
Rack block	Double-click	Open its editor or file chooser
Rack block	Drag sideways	Reorder (BLOCK NAM 2 and IR are fixed)
Rack block	Drop a file or folder	Load it into that block
Eye icon	Click	View that block with the shared controls
Chain icon	Click	Link / unlink that effect between L and R
NAM or IR list	Click a row	Load it
IR list	Click the selected row again	Deselect: clears the response and bypasses the IR block
NAM or IR list	Drop a file or folder	Load it
▲ / ▼	Click	Previous / next entry in the list
Dialog window	Esc	Close
Any control	Hover	Tooltip

22. Troubleshooting

Symptom	What to check
DI is silent	Press PLAY; returning from IN never resumes automatically. Check the loaded file, its gain and whether the playhead has reached its end
DI file rejected	Each file must be mono WAV, AIFF or FLAC; export each side separately from your DAW
No sound at all	POWER is on; MAIN route points at the outputs you monitor; the right input channel is ticked in Audio Setup; INPUT VU moves when you play
Sound, but no amp tone	A capture is loaded and BLOCK NAM 2 is lit; the chain is not entirely bypassed (a bypassed chain passes dry audio)
A NAM block will not switch on	Load a valid capture into that block and wait for the load to finish; a valid capture enables it automatically
IR block stays bypassed	Load a valid response, and check you have not clicked the selected row a second time (that deselects it)
Clicks or dropouts while playing	Raise the audio buffer, use the manufacturer ASIO driver on Windows, and avoid very small buffers with several NAM blocks running, especially PLUS in parallel
A folder appears empty	It must contain .nam files, or short valid .wav responses, directly inside the folder - unless DEEP SEARCH is ticked
Level does not match other NAM software	Check the interface reference in Audio Setup and whether Input Cali is on
A capture sounds too loud or too quiet	Check IR VOL (-12 dB by default) and Input Cali before changing INPUT GAIN
The session did not come back	The player saves on a **normal** exit; after a forced quit the previous session is restored instead
Models or IRs went missing	They are referenced by path. Put them back, or load a .bnpp preset, which embeds them
TONE3000 list is empty	Check the account, the network and - on Windows - WebView2; or use the external browser
macOS refuses the first launch	Right-click the application and choose **Open**
Linux AppImage will not start	chmod +x it and confirm the system has a working x86_64 audio stack

23. Technical specifications

Item	Value
Signal paths	2 independent (L / R)
NAM blocks	NORMAL: 2 per path. PLUS: 2 per lane, 2 lanes per path (up to 8 across L/R)
Cabinet IR	1 shared rack IR per path; PLUS adds up to 2 per lane (4 per path)
DI file player	Standalone only; 1 shared mono file or 2 mono files in SPLIT; WAV / AIFF / FLAC, automatic resampling
Effects	8 per path, 3 algorithms and 6 main parameters each; PITCH adds HARMONIZER
Rack positions	NORMAL: 11. PLUS replaces the two NAM tiles with one double-width NAM / IR tile at NAM 1; the general IR stays separate
NAM architectures	A1, A2 Standard, A2 Nano - detected automatically
Sample rates	Whatever the device offers, 44.1 - 192 kHz
Buffer size	Whatever the driver offers; the engine internally uses at least 32 samples and splits oversized host buffers
IR length	Up to 8192 samples, resampled to the device rate, never normalized
Tuner range	~65 - 700 Hz, ±5 cents centred
Auto Cal range	-24 … +24 dB
Start-up fade	60 ms from silence
NAM change fade	15 ms out, silence until the new capture is installed, 20 ms in (raised-cosine)
IR change fade	12 ms crossfade
Native window	1537 x 1023, responsive, minimum 50 %
Interface refresh	60 fps, entirely off the audio thread

24. Integrity, third-party code and licences

Compare downloaded ZIP hashes against SHA256SUMS.txt. Bcho NAM Player includes JUCE and Neural Amp Modeler Core; their notices are in THIRD_PARTY_NOTICES.md.

The application grants no rights over third-party NAM captures or impulse responses. Follow the licence supplied with each capture or response, including anything you download through TONE3000 or share inside a .bnpp preset.